

Sprint 0 Research Log

Project: Autonomous Smart Car with Vision-Guided Navigation & Mapping

Team: Vision & Control Systems Group

Date: January 30th, 2026

Areas of Research

Vision-Based Obstacle Detection

Field	Entry
Resource Title	Autonomous Navigation and Mapping in GPS-Denied Environments at Defence R&D Canada
Author/Source	Colin McManus, Paul Furgale, Braden Stenning and Tim Barfoot
Link	http://asrl.utias.utoronto.ca/~tdb/bib/collier_nato12.pdf
Type	Documentation
Summary	This paper overviews Canadian defence research on autonomous navigation and mapping using GPS, including vision-based conveying, 3D mapping, LiDAR teach-and-repeat, and UAV navigation.
Key Findings	Vision-based conveying uses SIFT (Scale-Invariant Feature Transform) features and adaptive models for leader tracking, enabling GPS-free operation; lidar-based place recognition improves robustness across lighting conditions.
Constraints/Limitations	Vision systems are sensitive to lighting changes and dust, while lidar performance degrades in feature-sparse environments.
Feasibility for Project	The vision-based tracking approach is feasible for structured environments but requires adaptation for uncontrolled settings.
Intended Use/Inspiration	Inspiration can be drawn from adaptive visual tracking methods to enhance the reliability of obstacle detection across varying conditions.
Investigated By	Joseph Toma

Field	Entry
Resource Title	Enhancing the Autonomous Mapping and Navigation Unit for Service Robots
Author/Source	Luiza G. Soares; Natan H. C. Rocha; Gabriel E. Soares; Gilmar P. Cruz Júnior; Gabriel Gomes; Gabriel M. de Oliveira
Link	https://ieeexplore.ieee.org/document/11066085

Type	Documentation
Summary	This paper presents an enhanced autonomous mapping and navigation unit (AM&NU) for service robots, developed using sensor-fusion SLAM that combines 3D LiDAR, IMU, wheel encoders, and visual data.
Key Findings	The work demonstrates the advantages of using the FAST-LIO2 algorithm for map generation and compares its performance with the visual odometry approach of RTAB-Map.
Constraints/Limitations	The challenge of operating in environments with limited GPS coverage and physical accessibility constraints.
Feasibility for Project	This system was successfully implemented on a ground robot and packaged in a compact form, indicating high feasibility in small robotic platforms.
Intended Use/Inspiration	The sensor-fusion approach is directly applicable to our project, which requires robust GPS-denied navigation for robots in controlled environments.
Investigated By	Joseph Toma

Embedded Vision Hardware Options

Field	Entry
Resource Title	Raspberry Pi Camera Connector & Hardware Guide (Module 3 Camera & Zero 2 W Board)
Author/Source	Raspberry Pi Foundation (Official Documentation)
Link	https://www.raspberrypi.com/documentation/accessories/camera.html#camera-module-3
Type	Documentation/Tutorial
Summary	This guide explains the hardware interface for Raspberry Pi camera modules, describes the 22-pin CSI connector on Raspberry Pi Zero series (including Zero 2 W), and details how to attach the camera's ribbon cable correctly to the board.

Key Findings	Raspberry Pi Zero 2 W uses a 22-pin CSI connector for camera modules requiring a specific Zero-series camera cable. The ribbon cable must be inserted with the correct orientation into the camera and board connectors. Camera Module 3 (IMX708) is compatible with the 22-pin CSI interface used on the Pi Zero series. The resource explains how to initialize the camera in software (via Picamera2) to start the camera pipeline and verify operation through image capture
Constraints/Limitations	The physical cable and connector must match (22-pin cable for Zero boards). After the hardware connection, the camera must be enabled in software (OS configuration)
Feasibility for Project	The camera module is very feasible for the project as the documentation shows the camera hardware connectivity for Raspberry Pi Zero boards directly.
Intended Use/Inspiration	Hardware setup reference for computer vision systems on Pi Zero 2 W.
Investigated By	Evan Romano

Path Planning Algorithms

Field	Entry
Resource Title	Path Planning Algorithms for Autonomous Mobile Robots
Author/Source	Mohammad Ali AskariHemmat
Link	https://spectrum.library.concordia.ca/id/eprint/984157/
Type	Documentation
Summary	Use of a Hybrid A* path planner for ground robots to generate real-time, drivable, and collision-free paths in unstructured environments, followed by a smoothing algorithm.
Key Findings	The Hybrid A* algorithm combines discrete search with a continuous action space using Dubins paths for kinematic feasibility, and a

	subsequent smoothing step optimizes the path for safety and smoothness.
Constraints/Limitations	This approach requires a precise pre-existing map and perfect localization; the performance degrades in highly dynamic environments due to overcorrection and replanning.
Feasibility for Project	The system is highly feasible as it was successfully implemented and tested on a physical robot, with the MATLAB toolbox available for practical use.
Intended Use/Inspiration	It serves as a practical reference for implementing a complete kinematic investigation strategy for robots in known environments.
Investigated By	Joseph Toma

Mapping Approaches for Ground Robots

Field	Entry
Resource Title	AprilTags Visual Fiducial System
Author/Source	AprilRobotics (GitHub repository)
Link	https://github.com/AprilRobotics/apriltag
Type	Open-Source Software/Documentation
Summary	AprilTags is a visual fiducial marker system designed for robotics research and applications, supporting detection and pose estimation of printed markers using computer vision methods.
Key Findings	AprilTags enable the fast detection of 2D fiducial markers in images. They support multiple tag families with different sizes and coding densities. The technology provides pose estimation given known tag size and calibrated camera parameters.
Constraints/Limitations	Requires appropriate tag printing and camera calibration for accurate pose estimation. Performance depends on image quality and tag size relative to camera resolution.
Feasibility for Project	Highly feasible; used widely in ground robot navigation, SLAM, docking, and localization tasks.

Intended Use/Inspiration	Fiducial-based localization, robot positioning, and mapping landmarks for autonomous navigation.
Investigated By	Evan Romano

Comparable Autonomous Robotics Projects

Field	Entry
Resource Title	Intelligent Ground Vehicle Competition (IGVC) 2025 AutoNav Challenge – Twistopher Design Report
Author/Source	University of Oklahoma – Sooner Competitive Robotics Team
Link	http://www.igvc.org/d2025.htm http://www.igvc.org/design/2025/7.pdf
Type	Documentation
Summary	This design report documents the development of “Twistopher,” an autonomous ground vehicle built for the 2025 Intelligent Ground Vehicle Competition (IGVC). The system is designed to navigate an obstacle-filled outdoor course using onboard perception, localization, and navigation without human intervention. The report covers system architecture, mechanical and electrical design, vision-based obstacle detection, localization, navigation logic, and safety systems.
Key Findings	The vehicle uses a modular software architecture that separates perception, localization, and navigation, enabling parallel development and testing. Vision-based obstacle detection is implemented using multiple cameras and OpenCV-based image processing to generate 2D obstacle maps. Navigation decisions are made using heuristic-based methods rather than full global path planning to reduce latency. Strong emphasis is placed on system safety, redundancy, and logging for debugging and performance evaluation.
Constraints/Limitations	The system relies on multiple cameras, high-performance onboard computing, and custom hardware, which increases cost and complexity beyond what is feasible for smaller projects. The approach is designed for outdoor environments and uses GPS and large-scale sensors that may not directly translate to small

	tabletop platforms. Some methods assume significant computational resources and extensive testing time.
Feasibility for Project	While the full system is beyond the scope of this project, several concepts are feasible and scalable. These include modular software organization, vision-based obstacle detection using camera data, simplified mapping approaches, and heuristic navigation strategies.
Intended Use/Inspiration	Inspiration for overall system architecture, separation of perception and navigation pipelines, vision-based obstacle detection workflows, and practical testing strategies.
Investigated By	Mohammed Alamen Qassab

Research Summary

Field	Entry
Common Design Constraints	Limited onboard computing resources, sensitivity of vision systems to lighting and environmental conditions, lack of access to high-end sensors such as LiDAR and GPS, and the need to operate reliably in a small-scale, controlled environment. Mechanical robustness, wiring reliability, and power safety are also recurring constraints across comparable systems.
Feasible Techniques Identified	Vision-based obstacle detection using OpenCV and a single camera, fiducial marker systems such as AprilTags or ArUco for localization cues, simplified mapping approaches based on local environment representation, heuristic or lightweight path planning methods, and modular software architectures that separate perception, planning, and control. Embedded vision platforms such as Raspberry Pi Zero 2 W are suitable for basic real-time processing.
Approaches to Avoid	Full-scale SLAM systems that require LiDAR, IMU fusion, or high-performance computing, reliance on GPS-based localization, complex global path planning algorithms with heavy computational requirements, and

	multi-camera or sensor-heavy setups that significantly increase system complexity and integration risk.
Impact on Final System Design	The final system will prioritize simplicity, modularity, and reliability by using a single camera-based perception pipeline, lightweight mapping and planning logic, and a scalable software architecture. Design decisions will focus on techniques proven effective in larger systems but adapted to fit the computational and mechanical constraints of a small autonomous smart car platform.

Informational Resources

- <https://www.flyability.com/blog/simultaneous-localization-and-mapping>
- <https://www.viam.com/post/building-a-map-for-your-autonomous-robot-heres-your-guide-to-slam-and-cartographer>